



NQRUST-MICROVM

Isolate Everything | Compromise Nothing

A production-ready, self-hosted Rust-powered platform for running lightweight microVMs, isolated Docker containers, and serverless functions — on your own hardware, from a single operator dashboard. Full data sovereignty, zero vendor lock-in.

125ms

VM Boot

3-in-1

VM · Container
· FaaS

KVM

Hardware
Isolation

0

Cloud
Lock-in

Version 2.0 - Brochure
June 2026

INTRODUCTION

What Is NQRust-MicroVM?

A self-hosted platform for running and managing lightweight virtual machines, Docker containers, and serverless functions on your own Linux hardware.

NQRust-MicroVM is a **production-ready** platform built on lightweight microVM technology with KVM hardware virtualization underneath — engineered for organizations that demand **hardware-level isolation** for every workload. Three Rust services and a web dashboard, installed in one command. The platform runs entirely on your infrastructure, whether on-premise or fully air-gapped — **complete data sovereignty** over workloads, data, and configurations.

What You Can Run



Virtual Machines

Isolated microVMs with their own kernel, rootfs, CPU/memory limits, network interface, and persistent storage volumes. Browser terminal, full & differential snapshots, and reusable templates.



Containers

Deploy Docker containers inside microVMs — full Docker API compatibility with an isolated kernel underneath. Container escape is structurally impossible.



Serverless Functions

Write and deploy Node.js, Python, or Ruby functions on-demand inside isolated VMs. In-browser code editor, interactive playground with live execution logs.

Four Services. One Platform.



Manager

Central orchestration server. VM lifecycle, image registry, networking, storage, RBAC, snapshots, and templates.



Host Agent

Runs on each KVM host. Drives the microVM hypervisor via Unix sockets. Multiple agents per Manager for horizontal scaling.



Guest Agent

Tiny binary auto-deployed inside every VM. Reports CPU, memory, network, and disk metrics — zero manual setup.



Operator Console

Web dashboard with browser terminal, real-time metric graphs, code editor, and container logs.



PLATFORM CAPABILITIES

Features That Drive Your Business

Every feature solves a real business problem — not just a technical checklist.



Ultra-Fast Boot

VMs boot in under 125 ms with as little as 5 MB overhead each. Spin up disposable environments at container speeds — with VM-grade isolation.



KVM Hardware Isolation

Every workload runs in its own isolated microVM with a dedicated kernel. Container-escape style attacks are structurally impossible.



Web Dashboard

Full operator UI in the browser. 7-tab VM detail view (overview, terminal, metrics, storage, network, snapshots, config) plus built-in shell terminal streamed over WebSocket — no SSH client setup.



Snapshots & Templates

Full and differential snapshots with instant restore. Save any VM configuration as a reusable template — deploy consistent environments in seconds.



Multi-User & RBAC

Three roles: Admin, User, Viewer. Per-resource ownership and per-user preferences — hand out the dashboard without handing out root.



Pluggable Storage

Mix local, NFS, iSCSI+LVM, TrueNAS, and SPDK backends. Add or remove storage from the UI without restarting the manager. Bring your own SAN or NAS — no migration required.



Networking & Clustering

Four network types: NAT, Isolated, Bridged, and VXLAN overlay for multi-host clustering. Built-in port forwarding to route external traffic to VM services. Scale across physical machines anytime.



Image Registry

Built-in registry for kernels, rootfs, and Docker images. Import from URL, local path, or DockerHub. Tarball upload for airgapped sites.



REST API + Live Metrics

Every action API-accessible with interactive Swagger UI at `/swagger-ui/`. Live CPU, memory, network, and disk metrics streamed over WebSocket for every VM and host. CI/CD-ready.

USE CASES

When Containers **Aren't Isolated Enough**

Built for teams whose workload model demands a real kernel boundary.



Government & Defense

Air-gapped deployment for classified environments. Full offline installation, **data sovereignty** on your infrastructure, isolated networking. RBAC ensures **complete audit trail** for regulatory compliance.



Agentic AI & Code Execution

Give LLM agents a real Linux machine — shell, filesystem, network — without giving them your host. **Per-job disposable VMs**, full Docker API for tool use, browser terminal for human-in-the-loop oversight.



On-Prem Multi-Tenant Platforms

Hand internal teams or customers self-service VMs with strict isolation, Admin/User/Viewer RBAC, and **pluggable storage that maps to your existing SAN**. Docker inside VMs = familiar workflow + structural escape protection.



Edge Computing & IoT

Minimal footprint for edge devices. **VXLAN overlay** for multi-site distributed deployment. Airgapped install for remote locations. Bridged networking for **direct device access** at operational sites.

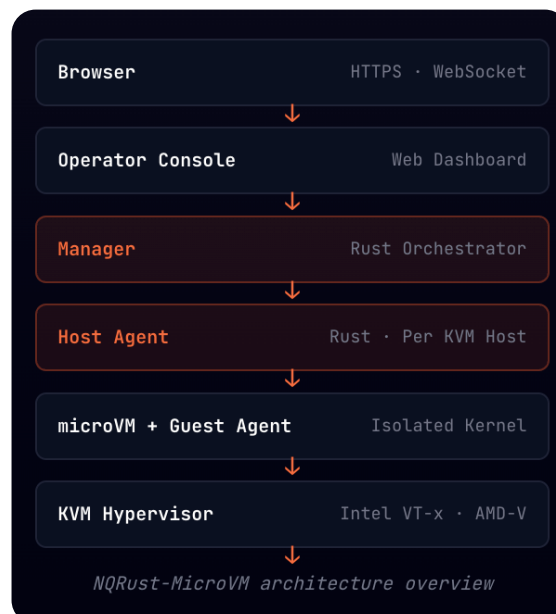
UNDER THE HOOD

Three Rust services. **One operator surface.**

The **Manager** is the central orchestration server handling VM lifecycle, image registry, networking, storage backends, users, and RBAC. It communicates with one or more **Host Agents** on each KVM host, which drive the microVM hypervisor via Unix sockets.

Inside every VM, a **Guest Agent** — a tiny static Rust binary deployed automatically — reports metrics directly to the Manager, providing **real-time visibility** across the entire fleet without manual configuration.

The multi-host architecture enables a single Manager to orchestrate many Agents across physical machines. **VXLAN overlay networking** allows transparent cross-host VM communication — ideal for distributed deployments at any scale.



WHY NQRUST - MICROVM

More Than a Platform. A Strategic Decision.

Security, independence, and operational efficiency — engineered for organizations that own their infrastructure.

Data Sovereignty & Full Control

Runs entirely on your servers. No tenant, no outbound calls, no vendor lock — not a single byte leaves your perimeter. Uncompromising compliance for government, defense, and enterprise environments.

Layered Security, Zero Compromise

Every workload in an isolated microVM with a dedicated kernel, protected by KVM hardware virtualization. True hardware-level tenant isolation you can prove to auditors.

One Platform, Three Runtimes

VMs, Docker containers, and serverless functions in one dashboard with one API. Dramatically reduces tool sprawl, training overhead, and operational complexity for your IT team.

Predictable Costs, Long-Term Value

Runs on hardware you already own. No surprise cloud bills, no egress fees, no vendor pricing changes. TUI installer gets you production-ready without consultants or multi-day training.

Ready to host your own cloud?

Explore the live demo, read the docs, or contact our sales team for a tailored quote and guided installation. Quotes returned within 48 hours.

► LANDING PAGE

<https://microvm.nexusquantum.id/>

► LIVE DEMO

<https://demo.microvm.nexusquantum.id/dashboard>

► DOCUMENTATION

<https://microvm.nexusquantum.id/docs/>